

1 §2.2 Separable Equations



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

Instructor: David Earn

Lecture 6
§2.2 Separable Equations
Friday 18 September 2026

Announcements

- The TA is Illya Kierkosz (kierkosi@mcmaster.ca)
 - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: <https://davidearn.github.io/math2c/>
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
 - Subject: Math 2C03: conflict with test window
- Office hour: Thursdays, 9:30–10:20 am, after class.

Logistic Model: exact solution

What can we learn from the exact solution?

$$P(t) = P_0 \frac{K}{P_0 + (K - P_0)e^{-rt}}$$

- If $P_0 = 0$ then $P(t) \equiv 0$ (*equilibrium*)
- If $P_0 = K$ then $P(t) \equiv K$ (*equilibrium*)
- If $P_0 > 0$ and $P_0 \neq K$ then as $t \rightarrow \infty$, $P(t) \rightarrow K$
 - $\therefore K$ is a *stable* equilibrium (and 0 is *unstable*)
- If $0 < P_0 < K$ then as $t \rightarrow -\infty$, $P(t) \rightarrow 0$
- If $P_0 > K$ then there is some time $t_\star < 0$ when the solution $P(t)$ is undefined at $t_\star$ (!) so there is a solution only for times $t \in (t_\star, \infty)$. Should we be concerned about this model?
- Is $P_0 > K$ reasonable biologically? Could a population start above the carrying capacity?

Separable Equations

The Malthusian and logistic models are both *autonomous* first order ODEs.

$$\frac{dP}{dt} = f(P) \quad \text{just like} \quad \frac{dy}{dt} = f(y)$$

Any nonconstant solution of such an ODE, on an interval where its right-hand side is nonzero, can be found at least implicitly *if* it is possible to find the anti-derivative $\int \frac{dy}{f(y)}$.

More generally, any *separable* first order ODE can be reduced to a problem in integration:

$$\frac{dy}{dx} = g(x)h(y) \quad \implies \quad \int \frac{dy}{h(y)} = \int g(x) dx$$

Separable Equations

Example (Separable IVP)

Solve $\frac{dy}{dx} = xy^2$ subject to $y(x_0) = y_0$

- Unique solution?

YES, for any initial condition.

$f(x, y) = xy^2$ and $\frac{\partial f}{\partial y}(x, y) = 2xy$ are continuous on $\mathbb{R}^2$.

- Can we separate and solve?

YES, provided $y \neq 0$, so we must first deal with initial condition $y(x_0) = 0$.

$y(x) \equiv 0$ is a solution, so it must be the unique solution for $y(x_0) = 0$ (for any $x_0 \in \mathbb{R}$).

Separable Equations

Example (Separable IVP [continued])

Solve $\frac{dy}{dx} = xy^2$ subject to $y(x_0) = y_0$

■ Now assume $y(x_0) \neq 0$. Then $\frac{1}{y^2} \frac{dy}{dx} = x$

$$\Rightarrow \int_{x_0}^x \frac{1}{y^2} \frac{dy}{dx} dx = \int_{x_0}^x x dx \Rightarrow \int_{y_0}^y \frac{dy}{y^2} = \int_{x_0}^x x dx$$

$$\Rightarrow -\frac{1}{y} \Big|_{y_0}^y = \frac{x^2}{2} \Big|_{x_0}^x \Rightarrow -\frac{1}{y} + \frac{1}{y_0} = \frac{x^2}{2} - \frac{x_0^2}{2}$$

$$\Rightarrow \frac{1}{y} = \frac{1}{y_0} - \frac{x^2 - x_0^2}{2} \Rightarrow y = \frac{1}{(x_0^2/2 + 1/y_0) - x^2/2}$$

Separable Equations

Example (Another Separable IVP)

Solve $\frac{dy}{dx} = xy(1 - y)$ subject to $y(x_0) = y_0$

- Can we separate and solve?

YES, provided $y(x_0) \neq 0$ and $y(x_0) \neq 1$.

- Is this IVP familiar?

YES, similar to the logistic model with $r = 1$ and $K = 1$.

- Now assume $y(x_0) \notin \{0, 1\}$. Then $\frac{1}{y(1 - y)} \frac{dy}{dx} = x$

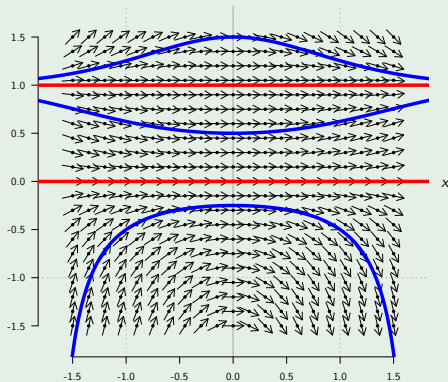
- Using ideas from the [previous example](#) and the [logistic model](#):

$$y = \frac{y_0}{y_0 + (1 - y_0)e^{-(x^2 - x_0^2)/2}}$$

Separable Equations

Example (Another Separable IVP [continued])

Solve $\frac{dy}{dx} = xy(1 - y)$ subject to $y(x_0) = y_0$



Note:

Slope of solution curves changes sign between equilibria.

Wasn't that supposed to be impossible??

What's going on???

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 2C03**
- Click on **Take Class Poll**
- Fill in poll **Sign changes between equilibria**
- .

Differential form of first order ODEs

First order ODEs are sometimes written in *differential form*,

$$A(x, y) dx - B(x, y) dy = 0, \quad (\heartsuit)$$

which *does not designate either variable as dependent*.

If y is regarded as a function of x then Eq. ($\heartsuit$) means the same as

$$A(x, y) - B(x, y) \frac{dy}{dx} = 0,$$

and a solution is a function $y(x)$.

If x is regarded as a function of y then Eq. ($\heartsuit$) means the same as

$$A(x, y) \frac{dx}{dy} - B(x, y) = 0,$$

and a solution is a function $x(y)$.

Eq. ($\heartsuit$) is *separable* iff it can be rearranged as $P(x) dx = Q(y) dy$.

Practice problems: separable equations

In [Trench, §2.2](#):

- Finding all solutions: Exercises 1–6
- Solutions and direction fields: Exercises 7–10
- Initial-value problems and solution graphs: Exercises 11–16
- More practice finding explicit solutions: Exercises 24–26
- Population-model applications: Exercises 27–29